Tender document - Components

L Abinaya (19MK01)
ME Infrastructure Engineering

THE PROCESS OF CONSTRUCTION PROJECT DEVELOPMENT
Client's intention to build
Feasibility study
Appoint consultants
Design and costing
Prepare tender Document
Tendering
Evaluate & Select contractor
Award tender and contracting
Construction
Completion and handover

Tender

- an offer to execute a specified work within a certain time under certain conditions of contract between the contractor and the client for a certain amount of money.
- a submission made by a prospective supplier in response to an invitation to tender. It makes an offer for the supply of goods or services.
- **Tendering** - the process of inviting bids for projects .
- **Objective -** to eliminate favoritism and corruption in awarding works to construction companies.

Types of tendering

- Open tendering
- Selective tendering
- Negotiated tendering

Open tendering

- Bidding process that is open to all qualified bidders
- usually published in the newspaper and internet
- chosen on the basis of price and quality.
- main tendering procedure employed by both the government and private sector.

Advantages

- It allows any interested contractor to tender. Therefore it gives opportunity for an unknown contractor to compete for the work.
- The tender list can be long as too many contractors tendering for one job.
- Allowing the tender list to be made without bias. Client will obtain the bargain possible. No favouritism in selecting contractors.

Disadvantages

- Inexperienced firms
- High aggregate cost for tender
- High evaluation time
- Unrealistic competition

Selective tendering

A short list of contractor is drawn up and they are invited to submit tenders.

Purpose of selective tendering

- to improve the quality of the bids received
- to ensure that contractors with the necessary experience and competence are given the opportunity
- for specific reasons of the employer, eg security reasons in government projects. etc.
- to make the tendering procedure more manageable

Advantages

- reduced cost of tendering
- reduced tender documentation
- shorter tender periods
- Better management of the tender process

Disadvantages

- Tender Price may be invariably higher than would have been in open tendering.
- Greater chance of collusion.
- Favoritisms may occur in the short listing.

Negotiated tendering

- single-stage tender carried out between the client and just one contractor.
- appropriate for highly specialised contracts (where there may be a limited number of potential suppliers), or for extending the scope of an existing contract.

Advantages

- Only reputed contractor are invited for negotiation.
- The Contractor can contribute his expertise during design stage.
- Early Commencement of work on site. It shortens the period involved in appointing the contractor.
- Reduce risk of failure
- Best alternative for the employer to adopt under special circumstances such as emergencies, security reasons etc.

Disadvantages

- The cost work is likely higher than competitive tender.
- Reduces the availability of work for other contractors.

Preparation of tender

The knowledge of the procedure adopted by the contractor in arriving at the figures he has quoted in his proposal is essential to all engineers as it would enable them to attain better results while preparing contract.

- Careful study of the contract documents
- Sub contraction work
- Visit to the site
- Time of completion
- Unremunerated work
- Estimation of an cost
- Final estimate for tendering

Components of tender document

Section I	Invitation to tenders
Section II	Instructions to Tenderers
Section III	Tender Data Sheet
Section IV	General Conditions of Contract (GCC)
Section V	Special Conditions of Contract (SCC)
Section VI	Specifications
Section VII	Drawings
Section VIII	Bill of Quantities
Section IX	Forms of Tender

Notice inviting tender

- a public document announcing a work and inviting interested parties to submit their bids.
- call for bids, call for tenders, invitation to tender (ITT), invitation for bid (IFB) or invitation to bid (ITB) or notice inviting tenders (NIT, a term often used in purchasing in India)
- a procedure for generating competing offers from different bidders
- An invitation to tender will include information describing the goods or services required in sufficient detail to enable the tenderer to prepare an accurate tender that is in a prescribed format so that it is easy to compare with other tenders.

An invitation to tender might include:

- Letter of invitation to tender
- Form of tender.
- Preliminaries (including pre-construction information and site waste management plan).
- Form of contract, contract conditions and amendments.
- Employer's information requirements.
- Tender pricing document
- Drawing schedule and Design drawings
- Instructions to tenderer explaining the tender process.
- The timescale for the tender process (including the address and time for the return of tenders).
- The evaluation process and any evaluation criteria.
- The submission required in response to the invitation to tender.

Tender No. & Date

Nature of Work

Minor Economics

Genuine Security Deposit

How to Pay

Total Duration

Time to Apply!

Time to Hurry!

Various Details & Links

Address Of Issuer

भारतीय कंटेनर निगम लिमिटेड
(भारत सरकार का उपक्रम - रेल मंत्रालय)
कॉनकोर CONCOR
CONTAINER CORPORATION OF INDIA LIMITED
(A Govt. India Undertaking- Ministry of Railways)

1st Floor, BPCL Bldg, 7 Chitnavis Marg, Civil Lines, Nagpur. Tel.: 2540406

TENDER NOTICE

CONCOR invites Sealed Tender for the following work:-

Tender No. & Date	CON/CR/NGP/801/2012/42 Dt. 19.03.2012
Description of Work	Tender for Hiring of Loaded Reach Stacker & Forklift/ Empty Container Handler at Nagpur Container Terminal.
Cost of Tender Document	₹ 3150/- (Rs. Three Thousand One Hundred Fifty only) including VAT @ 5%
Earnest Money Deposit	₹ 3,30,000/- (Rs. Three lakhs Thrity Thousand only)
Mode of Payment	Draft favoring "Container Corporation of India Ltd." Payable at Nagpur
Contract Period	4+1 Years
Date of Sale of Tender Document	From 19/03/2012 to 09/04/2012 during office hours in Regional Office.
Date of Tender Submission Before 15.00 Hrs on 10/04/2012	**Date of Opening of PQ bids On 10/04/2012 at 15.30**

For eligibility criteria and other details please log on to www.concorindia.com or contact at above address.

Chief General Manager

INDIAN INSTITUTE OF TECHNOLOGY, KANPUR
INSTITUTE WORKS DEPARTMENT, CENTRAL OFFICE

Notice Inviting e-Tenders

The Superintending Engineer, on behalf of Board of Governors, IIT Kanpur, invites online composite percentage rate tenders in two envelope system from registered contractors in appropriate class in CPWD/MES/BSNL/Railway/State PWDs and Central PSUs for each of the following works:

Sl No.	Name of work	Estimated Cost (in Rs.)	Earnest Money (in Rs.)	Period
1.	Construction of Hall of Residence for Boys No-XII (A&B) (SH: Civil & Electrical works)	A) Civil- 68,61,19,485/- B) Electrical 9,17,63,905/-	87,78,840/-	24 months
2.	Construction of Hall Res[illegible] Girls (Phase-II) after demolishing of Hall of Residence No. VI (Girl Hostel-II). (Room Block: Ground+ Seven Storied, Dinning & Facilities Block: Ground+Two Storied) (SH: Civil & Electrical Works).	[illegible] 25,19,05,226/- B) Electrical 3,11,49,468/-	38,30,550/-	24 months
3.	Construction of Multistoried Residential flats (48 Nos.) (SH: Civil & Electrical Works).	A) Civil- 24,18,41,666/- B) Electrical 2,39,11,125/-	36,57,530/-	20 months

PunjabColleges.com

SAMPLE

Last date and time of submission of bid **on 02.12.2013 upto 3.30 PM.** All details are available on website **www.iitk.ac.in/iwd/tenderhall.htm**, **www.tenderhome.com**. The bids can only be submitted online at **www.tenderwizard.com/IIT**.

No. IWD/CO/2013/1480 Dated: 06.11.2013

Superintending Engineer

MAHINDA WORLD CITY (JAIPUR) LIMITED, JAIPUR

INVITATIONS FOR BIDS (IFB)

Date: -22.07.2013
Bid No:- MWCJL/HR/NFRA/IT&ITes/BOUNDARY WALL/T-38

1. **MAHINDRA WORLD CITY (JAIPUR) LIMITED** is developing a SEZ and invites item rate Bids for the below mentioned works from the selected Bidders.

2. Tender document can be downloaded from our web site www.mahindrawoldcity.com/jaipur/Tenders.html,. Hard copies of the document can be taken from the below mentioned address by paying Rs. 2000/- in the form of Demand Draft/ Bankers cheque from Nationalised/ scheduled bank..

   Mahindra World City (Jaipur) Limited
   SEZ Project Office,
   Vill & Po – Kalwara near Mahapura
   Tehsil- Sanganer, Dist - Jaipur -302029
   Phone No: 0141-4007025 – 29
   Fax : 0141-4007030

3. Bids must be delivered to **Mahindra World City (Jaipur) Limited**, SEZ Project Office, Vill & Po – Kalwara near Mahapura, Tehsil- Sanganer, Dist - Jaipur -302029, on or before **15:00 Hours** on **27.07.2013** for Hard Copy submission. If the office happens to be closed on the date of receipt of the Bids as specified, the Bids will be received on the next working day at the same time and venue.

4. Other details can be seen in the Bidding documents.

TABLE - IFB 1

S. No.	Name of work	Cost of document (Rs.)	Period of completion
1	CONSTRUCTION OF BOUNDARY WALL OF 850 RMT AT IT&ITes AREA, FOR MWC JAIPUR	2000.00 *	Three (03) Months

*For those Parties who will take the hard copies of the document.
In case the bid document has been downloaded from website, the bid fee of Rs. 2000/- shall be paid by Demand draft/ Bankers cheque in favour of Mahindra world City (Jaipur) Ltd. Payable at Jaipur.

Seal of office

Instructions to bidders

- provides bidders with the general guidelines and formal rules governing the tender process.
- adds clarity and transparency in order to clarify bidders' questions prior to the beginning of the formal tender process.
- It is usually preferable to submit the financial and technical proposals in separate sealed envelopes; evaluation should be a two-stage process, with only the bidders that are qualified technically proceeding to the financial evaluation. This process should be outlined clearly in the instructions to bidders.

Sections of ITT

A. Introduction
B. Tendering Documents
C. Preparation of Tenders
    A. Mandatory requirements
    B. Technical evaluation
    C. Financial evaluation
D. Submission of Tenders
E. Opening and Evaluation of Tenders
F. Award of Contract

SECTION II: INSTRUCTIONS TO TENDERERS (ITT) 6
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1. Scope of Tender 9
2. Source of Funds 9
3. Eligible Tenderers 9
4. One Tender per Tenderer 11
5. Alternative Tenders by Tenderers 11
6. Cost of Tendering 11
7. Site Visit and Pre-Tender Meeting 11
B. Tendering Documents 12
8. Content of Tendering Documents 12
9. Clarification of Tendering Documents 13
10. Amendments of the Tendering Documents 14
C. Preparation of Tenders 14
11. Language of Tender 14
12. Documents Constituting the Tender 14
13. Documents Establishing Eligibility and Qualifications of the Tenderer 15
14. Lots Package 16
15. Form of Tender 16
16. Tender Prices 16
17. Tender Currencies 17
18. Tender Validity Period 17
19. Tender Security and Tender Securing Declaration 18
20. Format and Signing of Tender 19
D. Submission of Tenders 20
21. Sealing and Marking of Tenders 20
22. Deadline for Submission of Tenders 20
23. Late Tenders 21
24. Modification, Substitution and Withdrawal of Tenders 21

24. Modification, Substitution and Withdrawal of Tenders 21
E. Opening and Evaluation of Tenders 22
25. Opening of Tenders 22
26. Confidentiality 23
27. Clarification of Tenders 23
28. Preliminary Examination of Tenders 24
29. Correction of Errors 25
30. Conversion to Single Currency 25
31. Comparison of Tenders 25
32. National Preference 26
33. Determination of the Lowest Evaluated Tender 26
34. Post-qualification of Tenderer 26
F. Award of Contract 27
35. Criteria of Award 27
36. Clarifications 27
37. Lake victoria south water works development agency's Right to Accept any Tender and to Reject any or all Tenders 28
38. Procuring Entities Right to Vary Quantities at the Time of Award 28
39. Notification of Award 28
40. Signing of Contract 29
41. Performance Security 29
42. Advance Payment 30
43. Adjudicator 30
G. Review of Procurement Decisions 31
44. Right to Review 31
45. Time Limit on Review 31
46. Submission of Applications for Review by the Public Procurement Administrative Review Board 31
47. Decision by the Public Procurement Administrative Review Board 32
48. Appeal on the decision of the Review Board 32

Tender data sheet

- provides clarifications on the general information contained in the instructions to bidders, including: scheduling, submission deadlines, evaluation procedures, logistic support, regulations and so on.

- Bidders may be required to include in their technical proposal elements such as:
  - an understanding of local conditions
  - an understanding of the requirements of the contract
  - information on the equipment and technologies to be used
  - a schedule of activities to reach any performance targets
  - information on the experience and skills of key management and technical staff to be assigned
  - staffing and staff development plans.

- The Employer should specify in the Tender Data Sheet information and requirements specific to the circumstances of the Employer, the processing of the procurement, the applicable rules regarding tender prices and currency and the tender evaluation criteria that will apply to the tenders.

- In preparing this section , the following aspects should be checked

(a)Informatin that specifies and complements provisions of ITT
(b)Amendments and/or supplements, if any, to provisions of ITT, as necessitated by the circumstances of the specific procurement

Tender Data Sheet (TDS)

Instructions to Tenderers Clause Reference

TDS Reference Number	ITT Clause Number	Amendments of, and Supplements to, Clauses in the Instruction to Tenderers
		A. Introduction
1.	**1.1**	The Procuring Entity is Lake Victoria South Water Works Development Agency
2.	**1.1**	Name of Project is; **CONSTRUCTION OF KAPKISIARA WATER AND SANITATION PROJECT**
3.	**1.2**	The expected completion date of the works is: 90 days (3 months) from commencement date.
4.	**1.3**	The objective of the project is to increase accessibility both for sanitation and water within the school.
5.	**2.1**	Name of the Program is 'Water to schools' Financial Year: 2018/2019 *Description of Works under the contract:* a) Supply of all materials and construction of 12 – door VIP Latrine at the school b) Construction of rainwater harvesting system and supply and installation of 10000L plastic tank
6.	**2.2**	The loan/ credit number: Not Applicable.
7.	**5.1**	Alternative Tenders: Not allowed in this Tender
8.	**5.2**	Alternative time for completion: Not Applicable.
9.	**5.3**	Alternative technical designs: Not Applicable.
10.	**7.3**	Pre-Tender visit shall NOT take place. However, Bidders are strongly advised to visit the site and familiarise themselves with site conditions before preparing bid documents. Costs for such site visits shall be borne by the bidder.
11.	**7.5**	Any clarifications/addendum will be transmitted within 5 days from the date of pre-tender visit – N/A
	7.6	Non-attendance at the pre-tender meeting will not result in disqualification.

General conditions of contract (GCC)

- Contains broad aspects relating to the roles, rights, duties and responsibilities of the parties that are to interact with each other during the course of construction
- a very detailed set of instructions which has been developed over the years and tested by the courts.
- It covers most of the normal conditions that may arise during an ordinary job.
- These are generally applicable to all contracts irrespective of the nature of work, supplier type and other factors

GENERAL CONDITIONS OF CONTRACT (GCC)

Table of Clauses

1. Definitions
2. Application
3. Country of origin
4. Standards
5. Use of contract documents and information
6. Patent rights
7. Performance security
8. Inspection and tests
9. Packing
10. Delivery of goods
11. Transportation
12. Insurance
13. Distribution of dispatch documents
14. Incidental services
15. Spare Parts
16. Warranty
17. Payment
18. Prices
19. Modification of contract
20. Assignment
21. Sub contracts
22. Delays in suppliers' performance
23. Liquidated damages.
24. Termination for default
25. Force Majeure
26. Termination for insolvency
27. Termination for convenience
28. Resolution of disputes
29. Governing language
30. Applicable law
31. Notices
32. Taxes and duties

PART II

STANDARD GENERAL CONDITIONS OF CONTRACT
FOR USE IN CONNECTION WITH WORKS CONTRACTS

DEFINITIONS AND INTERPRETATION

1. (1) Definitions : In these Standard General Conditions of Contract, the following terms shall have the meaning assigned hereunder except where the context otherwise requires :

(a) "Railway" shall mean the President of the Republic of India or the Administrative Officers of the Railway or of the Successor Railway authorised to deal with any matters which these presents are concerned on his behalf.

(b) "General Manager" shall mean the Officer in-charge of the General Superintendence and Control of the Railway and shall also include the General Manager (Construction) and shall mean and include their successors, of the successor Railway.

(c) "Chief Engineer" shall mean the Officer in-charge of the Engineering Department of Railway and shall also include Chief Engineer (Construction), Chief Signal & Telecommunication Engineer, Chief Signal & Telecommunication Engineer (Construction), Chief Electrical Engineer, Chief Electrical Engineer (Construction) and shall mean & include their successors, of the Successor Railway.

(d) "Divisional Railway Manager" shall mean the Officer in-charge of a Division of the Railway and shall mean and include the Divisional Railway Manager of the Successor Railway.

(e) "Engineer" shall mean the Divisional Engineer or the Executive Engineer, Divisional Signal & Telecom Engineer, Divisional Electrical Engineer in executive charge of the works and shall include the superior officers of Open Line and Construction organisations on the Railway of the Engineering, Signal & Telecom and Electrical Departments, i.e. the Senior Divisional Engineer/Deputy Chief Engineer/Chief Engineer, Senior Divisional Signal & Telecom Engineer / Dy.Chief Signal & Telecom Engineer, Senior Divisional Electrical Engineer / Deputy Chief Electrical Engineer and shall mean & include the Engineers of the Successors Railway.

(f) "Engineer's Representative" shall mean the Assistant Engineer, Assistant Signal & Telecommunication Engineer and Assistant Electrical Engineer in direct charge of the works and shall include any Sr.Section/Junior Engineer of Civil Engineering/Signal and Telecommunication Engineering & Electrical Engineering Departments appointed by the Railway and shall mean and include the Engineer's Representative of the Successor Railway.

(g) "Contractor" shall mean the Person/Firm/Co-operative Society or Company whether incorporated or not who enters into the contract with the Railway and shall include their executors, administrators, successors and permitted assigns.

(h) "Contract" shall mean and include the Agreement of Work Order, the accepted Schedule of Rates or the Schedule or Rates of Railway modified by the tender percentage for items of works quantified, or not quantified, the Standard General Conditions of Contract, the Special Conditions of Contracts, if any; the Drawing, the Specifications, the Special Specifications, if any and Tender Forms, if any.

(i) "Works" shall mean the works to be executed in accordance with the contract.

Special Conditions of Contract (SCC) or Particular Conditions of contract

- It is an extension of information to be included or modified in GCC
- These conditions are specific to project, region or organisation
- It is a document issued by the owner to be attached to the GCC

Special conditions of contract contains:

- Identification of contract parties
- Identification of parties' representatives and limitation on authority
- Commencement and completion date for contract performance
- Amount of contract security if required
- Detailed information about the project requirements
- Payment schedule

When any portion of GCC is contradicting any provision of SCC, unless a different intention appears, then the provisions of SCC will override GCC

4.1.4. **Steel Plates:**

4.1.4.1 **Scrap & Excess Materials:**

The contractor will prepare the plate cutting diagram in such a way that minimum scrap is generated, also the cut plates should be used at a proper place to reduce the scrap. The cutting plate diagram shall be got approved by the Engineer-in-Charge.

4.1.4.2 **Serviceable Materials:**

a) Plates.

i) Both sides are greater than 1M.

ii) If any side is less than 1M but greater than 0.5M and the total area by equal or greater than 2M2.

b) Piping

All pipes measuring 2M and above in length shall be treated as serviceable materials provided they are in good and acceptable condition. Pipe in less than 2M length shall be treated as scrap.

4.1.4.3 The contractor shall submit, every month or alongwith R/A bill which ever is earlier, an account for all the materials issued to him by the owner in the proforma prescribed for this purpose by the Engineer-in-Charge. The contractor shall ensure that the wastage of steel plates, and pipes issued to him by the owner for the work shall be within the percentage given below at the theoretical consumption.

a) Steel plates.

i) Scrap (inclusive of non-recoverable wastage). : 5% (Five percent).

ii) Wastage (non-recoverable wastage) : 0.5% (Half percent).

b) Pipes & Tubes

i) Scrap (inclusive of non-recoverable wastage). : 3.50% (Three & half percent).

ii) Wastage (non-recoverable wastage) : 0.5% (Half percent).

All off cuts/scrap and unused plates, and piping etc. shall be the property of the owner and shall be returned by the contractor to the owner's stores at site at his own cost. In case, the contractor fails to do so then the recovery for such quantities not returned by the contractor shall be effected from the contractor's bill (s) or from any other dues of the contractor to the owner at rates indicated above.

ISPRL

APPENDIX-III TO SCC

TIME SCHEDULE

DESCRIPTION OF WORK	TIME FOR COMPLETION
Laying the Reinforced cement concrete road on hilly terrain at an elevation ranging from +70 meters MSL(mean sea level) to +80 meters, i.e., from flare area to watch tower no: 2&3 , spreading approximately 670 meters length Road as per the Scope of Work specified in the technical section of the document **(TENDER NO. ISPRL/VSP/RCC ROAD LAYING TO HILL TOP /T9/16 dated 28.12.2016)**	**45 days**

Notes :

a) Time for completion shall be reckoned from date of issue of Fax/ Letter of Acceptance , whichever is issued earlier.

b) The time indicated for completing all works in all respects including submission of all reports as per specifications, codes, drawings and instructions of Engineer-in-Charge.

c) It should be noted that the period completion of all works given above includes the time required for Mobilization at site, carrying out the works as per the requirements of Contract documents, demobilization, preparation of all reports in requisite quantities as mentioned in the Bidding Document, rectification's, if any, rework etc., complete in all respects to the entire satisfaction of Owner/ Engineer-in-Charge and direction of Engineer-in-charge.

(SIGNATURE OF BIDDER)

Tender Drawings

- A list of drawing showing the title, number of the drawing and revision number should be provided with the invitation to tender letter or with the specifications.
- Maps should show the proposed location for works, possible pipeline routes, crossings and so on.
- **purpose**- to describe the project in sufficient detail so that the price submitted by the contractor can be expected to be realistic.
- Drawings must show sufficient detail so that there is no significant change and subsequently no significant change of the cost.

THE PROJECT

FOR

RECONSTRUCTION OF QUAYWALL, SLIPWAY AND CFHC OFFICE BUILDING AT GALLE FISHERY HARBOR DAMAGED BY TSUNAMI DISASTER

LIST OF DRAWINGS	
GENERAL	
P/034-G-001	PROJECT INFORMATION
P/034-G-002	EXISTING CONDITIONS (1)
P/034-G-003	EXISTING CONDITIONS (2)
QUAY STRUCTURES	
P/034-Q-001	GENERAL PLAN OF RECONSTRUCTION QUAY STRUCTURES
P/034-Q-101	QUAY WALL ARRANGEMENT
P/034-Q-102	TYPICAL SECTION AND COPING DETAIL
P/034-Q-103	MISCELLANIOUNS DETAILS
P/034-Q-201	SLIPWAY PLAN AND LONGI. SECTION
P/034-Q-202	SLIPWAY TYPICAL SECTIONS
P/034-Q-203	CONCRETE BLOCKS DETAIL
P/034-Q-301	RETAIN WALL DETAIL
BUILDING WORKS	
P/034-B-001	SITE PLAN
P/034-B-002	DEMOLITION WORKS
P/034-B-101	OFFICE BILDINGS FINISH SCHEDULE
P/034-B-102	GROUND FLOOR PLAN
P/034-B-103	1ST FLOOR PLAN
P/034-B-104	ELEVATION (1)
P/034-B-105	ELEVATION (2) AND SECTION
P/034-B-201	WINCH HOUSE

Specifications

- The specification defines the standard of workmanship and materials required by the client for completion of the project.
- It may include a description of the works, site conditions, access to the site, site establishment, supply of materials and so on.
- The contractor is obliged to meet these specifications.
- must be precise and unambiguous in interpretation.
- A specification may set out how all the work may be done, or just certain aspects, leaving the rest to the contractor's experience

- Specificationss should be structured according to work packages mirroring the separation of the works into sub-contracts. This makes it easier for the contractor to price and so may result in a more accurate tender

- Specifications vary considerably depending on the stage to which the design has been developed, ranging from performance specifications (open specifications that require further design work to be carried out) to prescriptive specifications (closed specifications where the design is already complete)

Parameters	Values
Engine Type	Natural Gas Engine water/Ethylene Glycol cooled
Rotational speed	1,520 rpm
Compression ratio	13.5:1
Cylinders number	4
Displacement	1,984 cm^3
Type of Electric generator	cooled asynchronous generator
Active power	19.7 kW generator
Voltage	400 V, AC, three-phase
Frequency	50 Hz
Generator efficiency	0.94
CHP electrical efficiency	0.34
CHP heat recovery efficiency	0.58
Max. Thermal power output	36.1 kW
Heating water Flow rate	2.9 m^3/h
Supply temperature	70-90°C
Return temperature	60-70 °C
Maximum operating pressure	6 bar
Required methane number	> 80
Rated NG consumption	5.84 Nm^3/h
Fuels	Natural gas, H / L according to DVGW G 260,

Bill of quantities (BOQ)

- a document prepared by the quantity surveyor that provides project specific measured quantities of the items of work identified by the drawings and specifications in the tender documentation.
- itemizes all work to enable a contractor to price the work for which he or she is bidding.
- The quantities may be measured in number, area, volume, weight or time.
- Preparing a bill of quantities requires that the design is complete and a specification has been prepared.

- As the offer is built up of prescribed items, it is possible to compare both the overall price and individual items directly with other tenderers' offers, allowing a detailed assessment of which aspects of a tender may offer good or poor value. It provides a fair and accurate system for tendering

- Bills of quantities are normally only prepared on larger projects. On smaller projects, or for alteration work the contractor can be expected to measure their own quantities from drawings and schedules of work

Objectives of the Bill of Quantities

(a) to provide sufficient information on the quantities of works to be performed to enable tenders to be prepared efficiently and accurately
(b) to provide a priced Bill of Quantities for use in the periodic valuation of works executed.

ITEM NO.	DESCRIPTION	UNIT	QTY	UNIT COST MAT'L	UNIT COST LABOR	UNIT COST TOTAL	AMOUNT
2.00	D-4: 12.5mm thk. tempered clear glass door panel on top and bottom FD-100 aluminum framing with door threshold in powder coated white color (including all hardwares & acccessories)	set	1.00	38,000.00	2,000.00	40,000.00	40,000.00
3.00	D-5a: 44.5mm thk. Tanguile solid door w/ Matte Code: 9348 Light Oak Straight Laminate Finish	set	2.00	4,000.00	400.00	4,400.00	8,800.00
4.00	D-6a: PVC door with bottom louver & stainless steel metal stiffener both sides for door closer (plain white color)	set	2.00	3,000.00	400.00	3,400.00	6,800.00
5.00	D- 7, 44.5mmthk fully louvered swing wood door on solid kd. Frame	set	1.00	3,000.00	400.00	3,400.00	3,400.00
6.00	D-8: 44.5mm thk. standard steel plate FHC door on Ga. 16 RHS jambs	set	1.00	10,000.00	400.00	10,400.00	10,400.00
8.00	D-10: 19mm dia. Vertical solid bars spaced at every 100mm with 10mm x 360mm horizontal punche flat bars frame	set	1.00	15,000.00	2,500.00	17,500.00	17,500.00
	Others (please specify)						
	Window						
1.00	12.5mm thk. Tempered clear glass panel on concealed top & bottom aluminum "C" clip holder						
	GP- 1, 1290mmL x 2427mmH	panel	1.00	31,500.00		31,500.00	31,500.00
	GP- 1, 1096mmL x 2427mmH	panel	2.00	26,600.00		26,600.00	53,200.00
	GP- 2, 1099mmL x 2427mmH	panel	3.00	26,700.00		26,700.00	80,100.00
	Others (please specify)						
	Door jamb						
1.00	D-5a, Wood Door - KD wood jamb	l.m.	9.80	400.00	50.00	450.00	4,410.00

Forms of tender

The form of tender is prepared by the client or their consultants and completed and signed by the tenderer. It is a formal acknowledgement that the tenderer understands and accepts the terms of conditions of the tender documents and any other requirements that are stipulated.

- Technical Bid Form, Financial Bid Form , Contact Details Form, Form for Financial Capacity, Form of Contract Agreement

Form of tender contains the following

- Latest return time and date.
- Date, name and address.
- Tender reference number.

- The price for which they are to do the works and any adjustments prices that have been agreed should the scope of the work or period of time change
- The date until which the price remains valid.
- The period of time that is allocated for the works, or an actual completion date if required.
- Acceptance of terms and conditions.
- Confirmation of the country's law that will govern the contract
- Any qualifications that apply.
- Who will bear the costs of preparing the tender.
- Any alternative proposals
- Confirmation of the tender being genuine and bona fide.
- Signature and tenderer details.

ANNEXURE - I

_____ RAILWAY

TENDER FORM (First Sheet)

Tender No. _____

Name Of Work _____

To

The President of India

Acting through the _____

_____ Railway

I/We _____ have read the various conditions to tender attached hereto and agree to abide by the said conditions. I/We also agree to keep this tender open for acceptance for a period of _____ days from the date fixed for opening the same and in default thereof, I/We will be liable for forfeiture of my/our "Earnest Money". I/We offer to do the work for _____ Railway, at the rates quoted in the attached schedule and hereby bind myself/ourselves to complete the work in all respects within _____ months from the date of issue of letter of acceptance of the tender.

2. I/We also hereby agree to abide by the Indian Railways Standard General Conditions Of Contract, with all correction slips up-to-date and to carry out the work according to the Special Conditions of Contract and Specifications of materials and works as laid down by Railway in the annexed Special Conditions/Specifications, Schedule of Rates with all correction slips up-to-date for the present contract.

3. A sum of Rs. _____ is herewith forwarded as Earnest Money. Full value of the earnest Money shall stand forfeited without prejudice to any other right or remedies in case my/our Tender is accepted and if :

(a) I/We do not execute the contract documents within seven days after receipt of notice issued by the Railway that such documents are ready; and

(b) I/We do not commence the work within fifteen days after receipt of orders to that effect.

4. Until a formal agreement is prepared and executed, acceptance of this tender shall constitute a binding contract between us subject to modifications, as may be mutually agreed to between us and indicated in the letter of acceptance of my/our offer for this work.

Signature of Witnesses :

(1) _____

(2) _____

_____

Signature of Tenderer(s)

Date _____

Address of the Tenderer(s)

_____

FORM-10

INTEGRITY PACT

General

This Agreement (hereinafter called the Integrity Pact) is made on _____ day of the month of _____ between on one hand, **Telecommunications Consultants India Limited (TCIL)** acting Through, _____ (Hereinafter called the "BUYER" which expression shall mean and include, unless the context otherwise requires, his successors in office and assigns) of the First Part and _____ (Name of the company) represented by _____ /Authorised Signatory (Name & Designation of the officer) (hereinafter called the "BIDDER" /"SELLER" which expression shall mean and include, unless the context otherwise requires, his successors and permitted assigns) of the Second Part.

WHEREAS THE BUYER invites bid against Tender/RFO/EOI/RFP/RFQ for the_____ (Name of the Stores/Equipment/Services, Ref No: date) and the "BIDDER" /"SELLER" is willing to submit bid for the same and

WHEREAS the BIDDER is a Private Company/Public Company/Government Undertaking/Partnership Firm/Proprietorship Firm/Registered Export Agency, constituted in accordance with the relevant law in the matter and the BUYER is a Public Sector Undertaking.

NOW, THEREFORE

To avoid all forms of corruption by following a system that is fair, transparent and free from any influence/prejudiced dealings prior to, during and subsequent to the currency of the contract to be entered into with a view to :-

Enabling the BUYER to obtain the desired said stores/equipment/Services at a competitive price in conformity with the defined specifications by avoiding the high cost and the distortionary impact of corruption on public procurement, and

Reference

- Essays, UK. (November 2018). Types Of Tenders And Tendering Process Construction Essay. Retrieved from https://www.ukessays.com/essays/construction/types-of-tenders-and-tendering-process-construction-essay.php?vref=1
- https://www.designingbuildings.co.uk/wiki/Tender_documentation_for_construction_projects
- http://pppue.undp.2margraf.com/en/16_3.htm

Sample tender documents

- GCC https://cpwd.gov.in/AllCirculars.aspx?Type=24
- Tender document https://www.google.com/url?sa=t&source=web&rct=j&url=http://ipindia.nic.in/writereaddata/Portal/Tender/164_1/1_Tendernotice_as_uploaded_in_CPP_Portal_on_20171213.pdf&ved=2ahUKEwiX1rjqk-_oAhWxzzgGHTZoCnIQFjAAegQIAxAB&usg=AOvVaw0S7e5f1mZtpwP2OGKJFKVr
- https://www.tcil-india.com/tender_details.php?tid=20/C/658&type=NIT

Thankyou